

Is Java Dead?

Java has been one of the most in-demand languages for over 20 years now. It is widely used across the industry and is often a compulsory part of the syllabus in most computer science courses across the world. But many developers are beginning to see it as an aged and primitive programming language, which, they predict, will soon become obsolete. This article tries to get to the heart of this issue by analysing popular developer surveys.

In the last week of December 2021, I received a call from a friend of mine. She is a programmer who just completed her master's degree in computer science. She was setting her learning goals for the upcoming year and asked me to recommend five programming languages that she should focus on. One of the recommendations I gave was to learn Java. "But Mir," she said, "Isn't Java dead?" I was taken aback by this statement. How can someone claim that Java is dead? It is one of the most widely adopted programming languages in the industry. It is also very much in use and in demand in the current market.

Later that day, I Googled 'Is Java dead?' just to see if there was any meat in this argument. This took me down a rabbit hole filled with a great many developers who do, in fact, believe that Java is mostly a redundant language with not much use in the future. Amongst the critics of Java were some great names in the industry such as Linus Torvalds. "What a horrible language," said Linus in an interview when asked about Java. That got my attention and I started to dig deeper. I noticed a pattern in the arguments that the developers shared against Java. The vast majority of Java critics seemed to have issues with the way this language is structured.

The best way to understand how the developer industry perceives a technology is to follow the data published on popular developer



survey reports. In this article, I have researched data from GitHub Octoverse Report 2021, JetBrains State of Developer Ecosystem 2021 and StackOverflow Developer Survey 2021. The insights that I drew from these reports hold the key to understanding what has gone right or wrong with the Java ecosystem.

A brief overview of Java

My assumption in this article is that the reader already has some notion of Java programming and how it functions. As such, I shall not go into too many details about the history of Java. Doing so will take us off topic. But it is always a good idea to share the basics so that we can brush up on what we already know about Java. Feel free to skip this section if you have a good understanding of the language.

Java is a high level, class based, object-oriented programming language. Built on the write once, run anywhere (WORA) model, it was developed by Sun Microsystems in 1995. Oracle acquired Sun Microsystems in 2010 and took charge of the development process of Java.

From 2010 onwards, the development of Java sped up under Oracle with new versions being released almost every six months. The latest version at the time of writing this article is Java 17, released in October 2021.

Design principles of Java

Before heading into the analysis of the developer surveys, I find it necessary from the context of this article to share with you Oracle's vision for Java. Listed below are the key points from the design principles of Java, as mentioned on the official website of Oracle.

Java is designed to be:

1. Simple, object-oriented and familiar
2. Robust and secure
3. Architecture-neutral and portable
4. High performance
5. Interpreted, threaded and dynamic

These points are about to become very relevant to this article. I shall be connecting the dots in the last section. For now, let us begin the analysis of the developer surveys.

GitHub Octoverse Report 2021

Every year, GitHub publishes a survey of its own platform to gain insights on developer trends. It is called the GitHub Octoverse Report. The following are the key insights from the 2021 report with respect to Java, as shown in Figure 1.

Java ranked second in the top ten programming languages used in the GitHub repositories from 2014 to 2018. JavaScript remained the most popular programming language.